FINAL PROJECT

October 25, 2026

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Dataset from: Travel Insurance Prediction Data: https://www.kaggle.com/datasets/tejashvi14/travel-insurance-prediction-data/code

Travel insurance can protect travelers from financial losses they might incur while traveling. Different plans cover risks ranging from last-minute flight cancellations to medical emergencies.

In this project, we will find out if there is a correlation between someone's travel history, income, and their decision to buy travel insurance based on the dataset provided.

[1]:
```
import pandas as pd
df = pd.read_csv("TravelInsurancePrediction.csv")
df.head()
```

[1]:

	Unnamed: 0	Age	Employment Type	GraduateOrNot	AnnualIncome	FamilyMembers	ChronicDiseases	FrequentFlyer	EverTravelledAbroad	TravelInsurance
0	0	31	Government Sector	Yes	400000	6	1	No	No	0
1	1	31	Private Sector/Self Employed	Yes	1250000	7	0	No	No	0
2	2	34	Private Sector/Self Employed	Yes	500000	4	1	No	No	1
3	3	28	Private Sector/Self Employed	Yes	700000	3	1	No	No	0
4	4	28	Private Sector/Self Employed	Yes	700000	8	1	Yes	No	0

[2]:
```
df.shape
```

[2]:
```
(1987, 10)
```

[3]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1987 entries, 0 to 1986
Data columns (total 10 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Unnamed: 0           1987 non-null   int64
 1   Age                  1987 non-null   int64
 2   Employment Type      1987 non-null   str
 3   GraduateOrNot        1987 non-null   str
 4   AnnualIncome         1987 non-null   int64
 5   FamilyMembers        1987 non-null   int64
 6   ChronicDiseases      1987 non-null   int64
 7   FrequentFlyer        1987 non-null   str
 8   EverTravelledAbroad  1987 non-null   str
 9   TravelInsurance      1987 non-null   int64
dtypes: int64(6), str(4)
memory usage: 217.6 KB
```

[4]:
```
df.describe()
```

[4]:

	Unnamed: 0	Age	AnnualIncome	FamilyMembers	ChronicDiseases	TravelInsurance
count	1987.000000	1987.000000	1.987000e+03	1987.000000	1987.000000	1987.000000
mean	993.000000	29.650226	9.327630e+05	4.752894	0.277806	0.357323
std	573.741812	2.913308	3.768557e+05	1.609650	0.448030	0.479332
min	0.000000	25.000000	3.000000e+05	2.000000	0.000000	0.000000
25%	496.500000	28.000000	6.000000e+05	4.000000	0.000000	0.000000
50%	993.000000	29.000000	9.000000e+05	5.000000	0.000000	0.000000
75%	1489.500000	32.000000	1.250000e+06	6.000000	1.000000	1.000000
max	1986.000000	35.000000	1.800000e+06	9.000000	1.000000	1.000000

Firstly, we run the four essential pandas DataFrame commands to inspect the TravelInsurancePredictions.csv dataset. Shape would help identify the size of the dataset, info helps identify any missing values or any wrong data types, head to provide a preview of the first 5 rows of the dataset, and last but not least, describe will help us identify any outliers, scale differences, and distributions.

[5]:
```
df.isnull().sum()
```

[5]:
```
Unnamed: 0             0
Age                    0
Employment Type        0
GraduateOrNot          0
AnnualIncome           0
FamilyMembers          0
ChronicDiseases        0
FrequentFlyer          0
EverTravelledAbroad    0
TravelInsurance        0
dtype: int64
```

No missing values in this dataset.

[1]:
```
import pandas as pd
from scipy.stats import pearsonr

df = pd.read_csv('TravelInsurancePrediction.csv')

df['EverTravelledAbroad_Num'] = df['EverTravelledAbroad'].apply(
    lambda x: 1 if str(x).strip().lower() == 'yes' else 0
)
df['FrequentFlyer_Num'] = df['FrequentFlyer'].apply(
    lambda x: 1 if str(x).strip().lower() == 'yes' else 0
)

r_abroad, p_abroad = pearsonr(
    df['EverTravelledAbroad_Num'], df['TravelInsurance']
)
r_flyer, p_flyer = pearsonr(df['FrequentFlyer_Num'], df['TravelInsurance'])

print(f"EverTravelledAbroad Pearson's r: {r_abroad:.4f} (p-value: {p_abroad})")
print(f"FrequentFlyer Pearson's r: {r_flyer:.4f} (p-value: {p_flyer})")
```

```
EverTravelledAbroad Pearson's r: 0.4332 (p-value: 1.0889790056484136e-91)
FrequentFlyer Pearson's r: 0.2321 (p-value: 1.0325506803560456e-25)
```

Using Pearson's correlations r, we measure the relationship between travel history and whether or not they bought travel insurance. For those who have traveled before, we find that r = 0.4332, and p = 1.09 x 10^91.

There is a moderate positive correlation, meaning customers who have traveled before are more likely to purchase travel insurance.

The p-value is almost zero, which means the relationship is significantly strong and not just random.

[2]:
```
r_income, p_income = pearsonr(df['AnnualIncome'], df['TravelInsurance'])

print(f"Pearson's r: {r_income:.4f}")
print(f"P-value: {p_income:.4e}")
```

```
Pearson's r: 0.3968
P-value: 6.6350e-76
```

Now looking at the income and the decision to buy travel insurance, we find that r = 0.3968 and p-value is 6.64 x 10^-76.

There is a strong positive relationship between annual income and the decision to buy travel insurance. As the customer's income grows, they are more likely to purchase travel insurance. Also, the p-value is small, which indicates that the relationship is significantly high.

In comparison with travel history, the income's correlation is super close in strength to travel history, financial standing one of the two most dominant factors in purcchasing travel insurance.

[5]:
```
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

# 1. Load dataset
df = pd.read_csv('TravelInsurancePrediction (1).csv')
if 'Unnamed: 0' in df.columns:
  df = df.drop(columns=['Unnamed: 0'])

# 2. Preprocessing: Handle missing values and convert Yes/No columns to numeric
df = df.dropna()

for col in ['FrequentFlyer', 'EverTravelledAbroad', 'GraduateOrNot']:
  if col in df.columns:
    df[col] = df[col].apply(lambda x: 1 if str(x).strip().lower() == 'yes' else␣
 ↪0)

if 'Employment Type' in df.columns:
  df['Employment Type'] = df['Employment Type'].astype('category').cat.codes

# Create income brackets to group AnnualIncome alongside EverTravelledAbroad
df['Income_Bracket'] = pd.qcut(
    df['AnnualIncome'],
    q=3,
    labels=['Lower Income', 'Medium Income', 'Higher Income'],
)

# 3. Data Visualization: Multi-variable Bar Chart
bar_data = (
    df.groupby(['Income_Bracket', 'EverTravelledAbroad'])['TravelInsurance']
    .mean()
```

```
    .reset_index()
)
bar_data['EverTravelledAbroad'] = bar_data['EverTravelledAbroad'].map(
    {1: 'Yes', 0: 'No'}
)

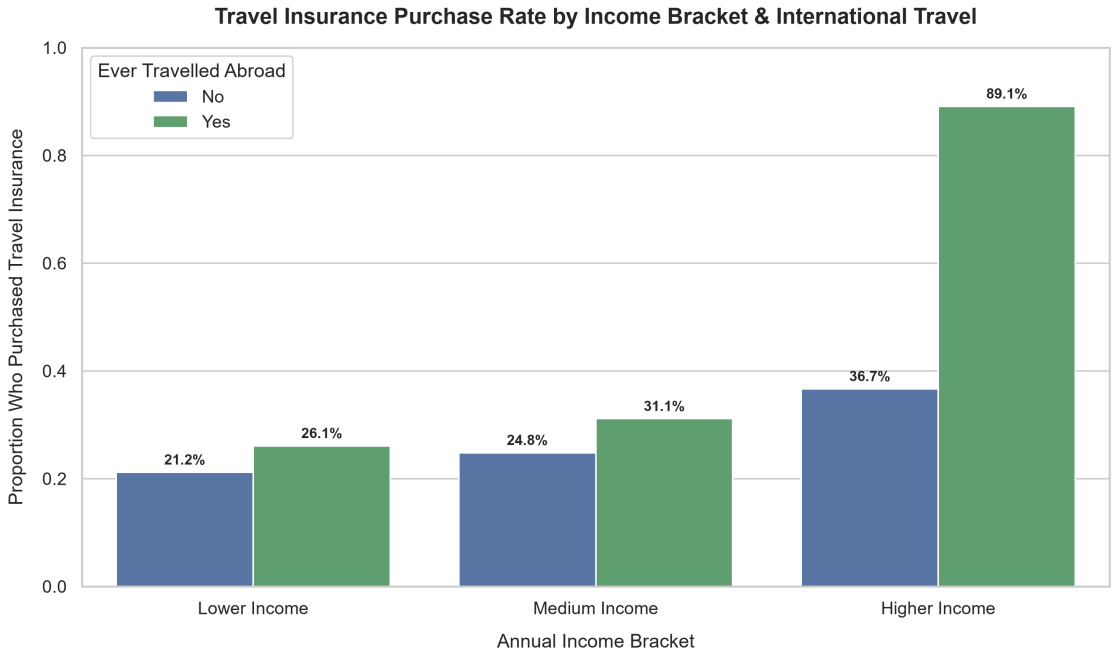
sns.set_theme(style='whitegrid')
plt.figure(figsize=(10, 6), dpi=300)

ax = sns.barplot(
    x='Income_Bracket',
    y='TravelInsurance',
    hue='EverTravelledAbroad',
    data=bar_data,
    palette=['#4c72b0', '#55a868'],
)

plt.title(
    'Travel Insurance Purchase Rate by Income Bracket & International Travel',
    fontsize=14,
    fontweight='bold',
    pad=15,
)
plt.xlabel('Annual Income Bracket', fontsize=12, labelpad=10)
plt.ylabel('Proportion Who Purchased Travel Insurance', fontsize=12,␣
 ↪labelpad=10)
plt.ylim(0, 1.0)
plt.legend(title='Ever Travelled Abroad', loc='upper left')

# Add data labels on top of bars
for p in ax.patches:
  height = p.get_height()
  if height > 0:
    ax.annotate(
        f'{height:.1%}',
        (p.get_x() + p.get_width() / 2.0, height),
        ha='center',
        va='bottom',
        fontsize=9,
        fontweight='bold',
        xytext=(0, 3),
        textcoords='offset points',
    )

plt.tight_layout()
plt.savefig('income_abroad_insurance_bar.png', dpi=300)
plt.show()
```



DISCLOSURE: Utilized AI to generate the above.

PROMPT: You are helping with data analysis for CSV file TravelInsurancePrediction.csv dataset. Write Python using pandas, matplotlib, and seaborn:

(1) My target variable to analyze are [EverTraveledAbroad], [AnnualIncome] and [TravelInsurance]

(2) Preprocess the dataset to handle any missing values, convert categorical/binary columns (like Yes/No) into numeric formats.

(3) Data Visualization: Generate a clean, well-labeled bar chart.

This shows us that travelers on the higher income group and have traveled previously have purchased travel insurance at a very high rate at 89.1%.

[6]:
```
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

df = pd.read_csv('TravelInsurancePrediction (1).csv')
if 'Unnamed: 0' in df.columns:
  df = df.drop(columns=['Unnamed: 0'])

df = df.dropna()
for col in ['FrequentFlyer', 'EverTravelledAbroad', 'GraduateOrNot']:
  if col in df.columns:
```

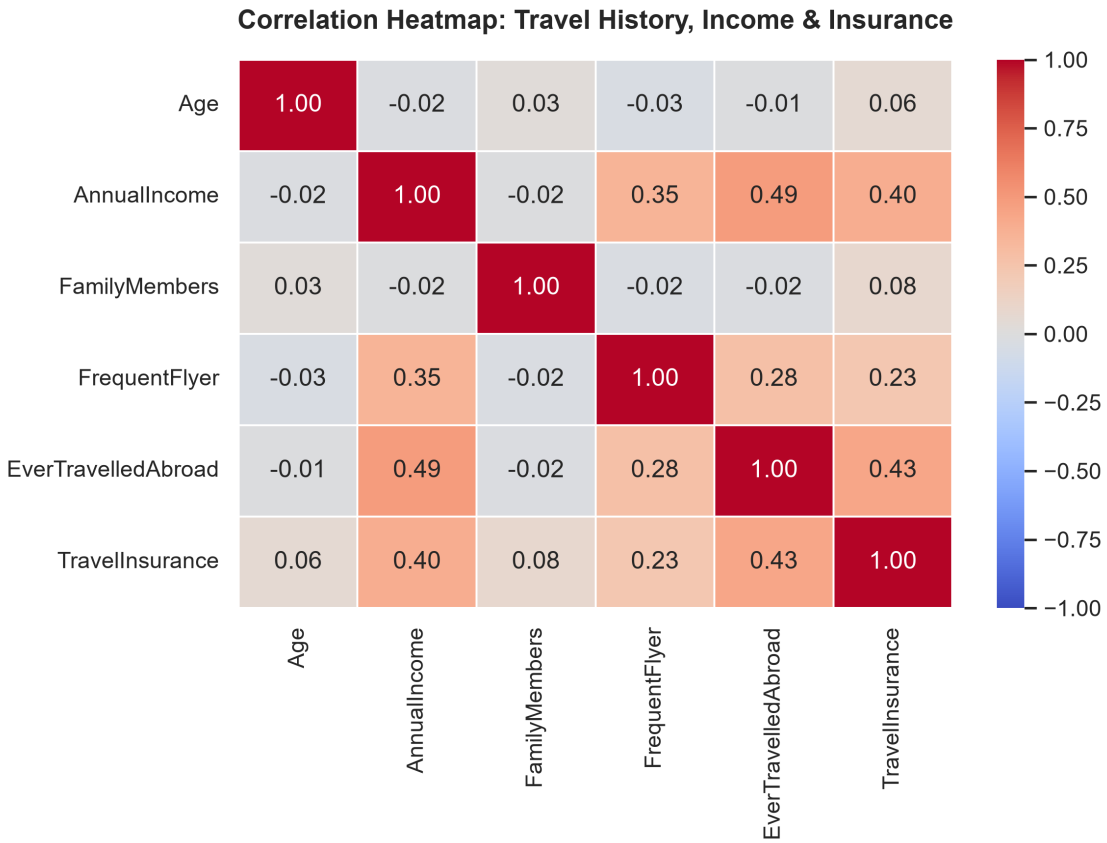
```
    df[col] = df[col].apply(lambda x: 1 if str(x).strip().lower() == 'yes' else␣
 ↪0)

selected_cols = [
    'Age',
    'AnnualIncome',
    'FamilyMembers',
    'FrequentFlyer',
    'EverTravelledAbroad',
    'TravelInsurance',
]
corr_subset = df[selected_cols].corr()

sns.set_theme(style='whitegrid')
plt.figure(figsize=(8, 6), dpi=300)

sns.heatmap(
    corr_subset,
    annot=True,
    fmt='.2f',
    cmap='coolwarm',
    linewidths=0.5,
    cbar=True,
    vmin=-1,
    vmax=1,
)

plt.title(
    'Correlation Heatmap: Travel History, Income & Insurance',
    fontsize=13,
    fontweight='bold',
    pad=15,
)
plt.tight_layout()
plt.savefig('subset_correlation_heatmap.png', dpi=300)
plt.show()
```



Above is the correlation heatmap for travel history, income, and travel insurance.

EverTraveledAbroad and TravelInsurance show the strongest direct positive correlation with purchasing insurance. AnnualIncome and TravelInsurance have an almost equally strong positive relationship.

[8]:
```
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

df = pd.read_csv('TravelInsurancePrediction.csv')
if 'Unnamed: 0' in df.columns:
  df = df.drop(columns=['Unnamed: 0'])

df['TravelInsurance_Label'] = df['TravelInsurance'].map(
    {1: 'Purchased Insurance', 0: 'No Insurance'}
)

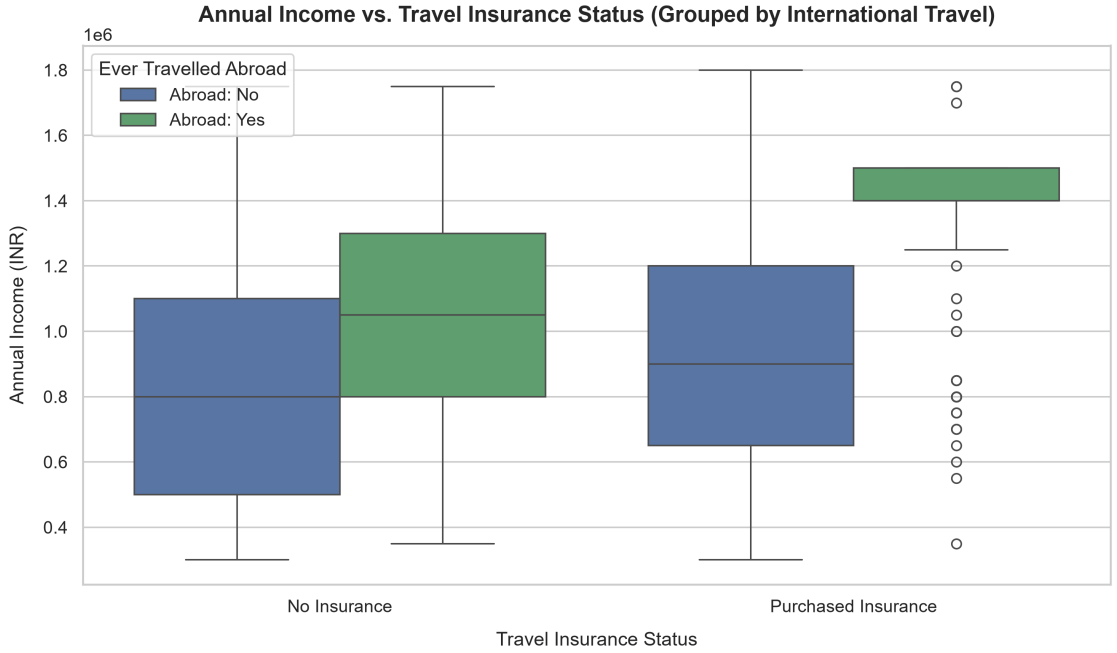
df['EverTravelledAbroad_Label'] = df['EverTravelledAbroad'].apply(
    lambda x: 'Abroad: Yes' if str(x).strip().lower() == 'yes' else 'Abroad: No'
)
```

```
sns.set_theme(style='whitegrid')
plt.figure(figsize=(10, 6), dpi=300)

sns.boxplot(
    x='TravelInsurance_Label',
    y='AnnualIncome',
    hue='EverTravelledAbroad_Label',
    data=df,
    palette=['#4c72b0', '#55a868'],
)

plt.title(
    'Annual Income vs. Travel Insurance Status (Grouped by International'
    ' Travel)',
    fontsize=14,
    fontweight='bold',
    pad=15,
)
plt.xlabel('Travel Insurance Status', fontsize=12, labelpad=10)
plt.ylabel('Annual Income (INR)', fontsize=12, labelpad=10)
plt.legend(title='Ever Travelled Abroad', loc='upper left')

plt.tight_layout()
plt.savefig('income_vs_insurance_hue_abroad_clean.png', dpi=300)
plt.show()
```



For the third chart, this boxplot compares the distribution of AnnualIncome between customers who purchased travel insurance and those who did not.

Each insurance status category is further split into two colored boxes representing international travel history:

Blue Box: Customers who have traveled internationally. Green Box: Customers who have never traveled internationally.

In conclusion, annual income and travel history are major behavioral indicators for purchasing travel insurance. Those travelers, especially those who have traveled abroad before, are three times more likely to purchase than those who have not traveled abroad before, making it a critical feature for both statistical insights and predictive modeling.

[ ]: